

Class 12 Mathematics

ICSE and CBSE Board Syllabus

ICSE Syllabus

Theme 1: Relations and Functions

- Types of relations and functions, domain and range.
- One-one, onto, inverse functions and composition of functions.

Theme 2: Algebra

- Matrices and determinants: operations, properties, minors, cofactors, inverse of a matrix.
- Solution of simultaneous linear equations using matrices.
- Continuity and differentiability basics.
- Applications of derivatives including tangents, normals, maxima and minima.

Theme 3: Calculus

- Indefinite and definite integrals and their properties.
- Applications of integrals as area under simple curves.
- Differential equations of first order and first degree.

Theme 4: Probability

- Conditional probability, multiplication theorem, Bayes' theorem, random variables and probability distributions.

Theme 5: Vectors and Three Dimensional Geometry

- Vectors, scalar/dot product, vector/cross product basics.
- Direction cosines, direction ratios, line and plane basics in 3D geometry.

CBSE Syllabus

Unit 1: Relations and Functions

- Relations, types of functions, inverse trigonometric functions and properties.

Unit 2: Algebra

- Matrices and determinants with applications to solving systems of equations.

Unit 3: Calculus

- Continuity and differentiability.
- Application of derivatives.
- Integrals: indefinite and definite integrals, fundamental theorem, applications.
- Differential equations of first order and first degree.

Unit 4: Vectors and Three-Dimensional Geometry

- Vectors, scalar product, vector product.
- Equation of line and plane in space.

Unit 5: Linear Programming

- Formulation and graphical solution of linear programming problems.

Unit 6: Probability

- Conditional probability, Bayes' theorem, random variable and probability distributions.